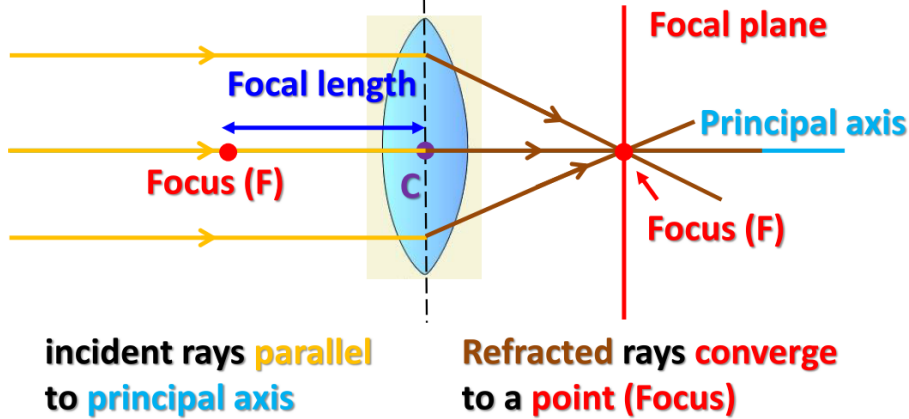
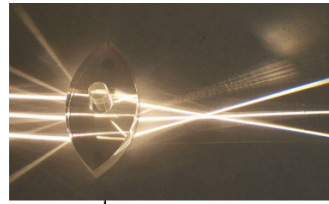


Convex lens

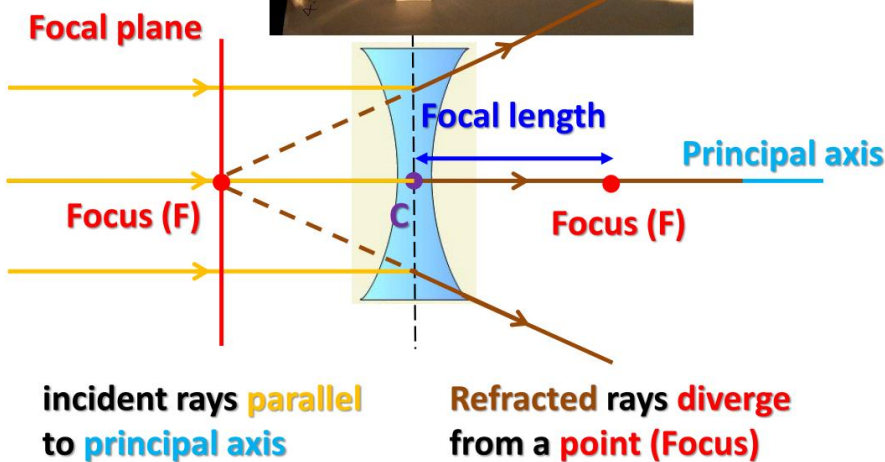
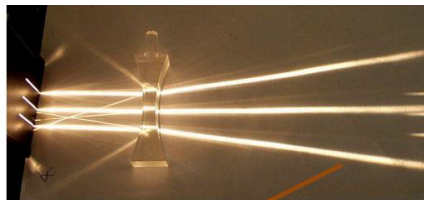


Convex lens = **Converging** lens

Symbol for
convex lens



Concave lens

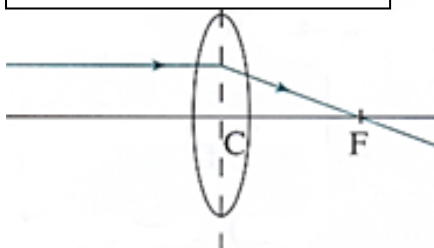


Concave lens = **Diverging** lens

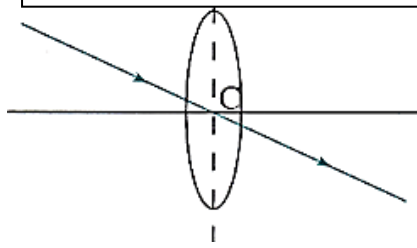
Symbol for
concave lens



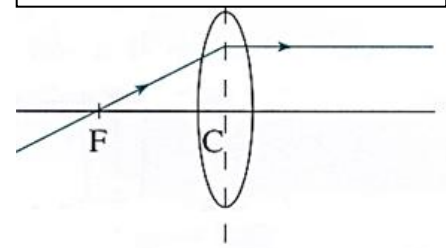
平衡入, Focus(F) 出



C 入 C 出, 直入直出

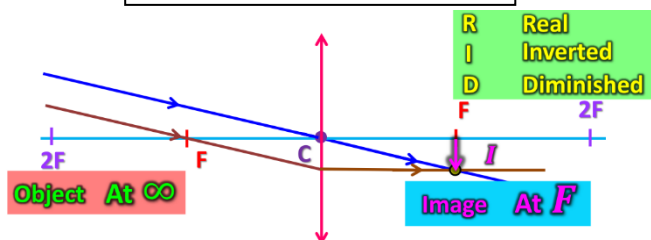


Focus(F) 入, 平衡出

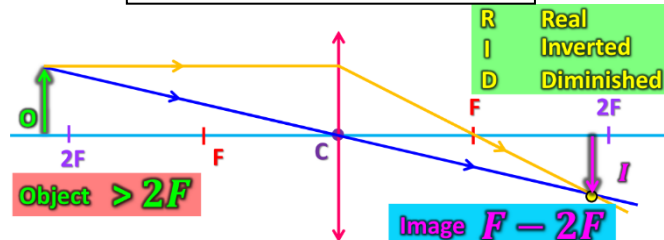


Construction of images by ray diagrams (i) Convex lens

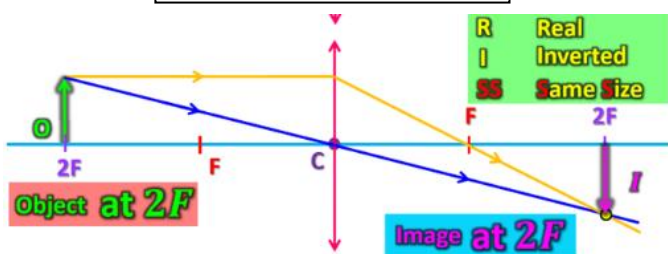
1. Object placed at infinity



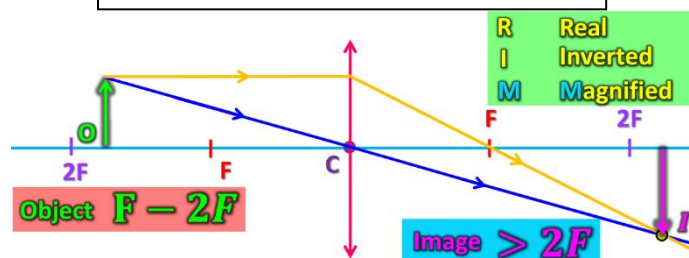
2. Object placed beyond $2F$



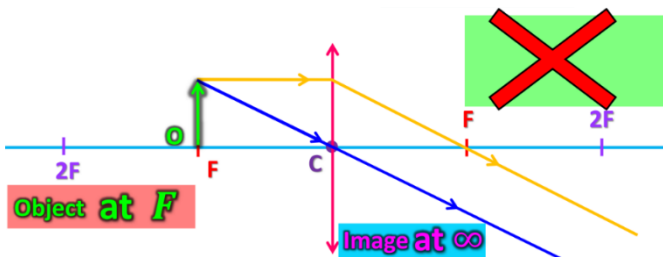
3. Object placed at $2F$



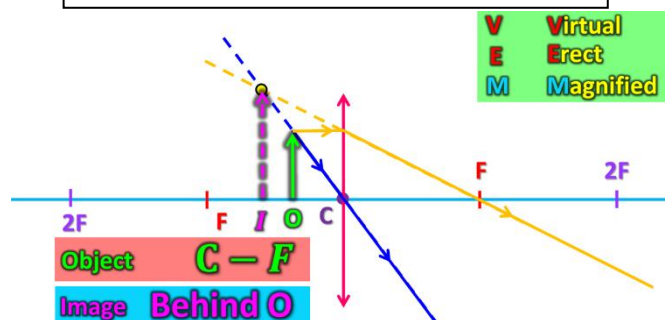
4. Object placed between F and $2F$



5. Object placed at focus F



6. Object placed between lens and F



Remarks:

- Convex lens can only give same size image when the object is placed at $2F$. Separation between the object and the image is then equal to 4 times of focal length, which is the minimum separation between object and the real image.
- Light is reversible and hence the position of the object and the real image can be interchanged.
- A lens with a larger curvature or greater refractive index bends light more and hence it has shorter focal length
- When part of lens is covered, a **complete** but **dimmer** (lower brightness) image is formed since some light rays are blocked and amount of light forming the image decrease.
- When two convex lenses of the **same size but different focal length f** are placed at the same distance from an object, same amount of light is distributed over the image in each case so that **smaller image** is **brighter** but **larger image** is **dimmer**.
- **Application** of convex lens:
 - (i) Magnifying glass
 - (ii) Spectacles for correcting long-sighted eye

1. Which of the following optical devices can form a virtual and magnified image?
- (1) Convex lens
 - (2) Concave lens
 - (3) Plane mirror
- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only
2. Which of the following optical devices can form a real and magnified image?
- (1) Concave lens
 - (2) Convex lens
 - (3) Plane mirror
- A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only
3. Parallel rays from a distant object converge on passing through a lens. Which of the following are correct?
- (1) The lens is convex.
 - (2) The image formed is real.
 - (3) Image distance is equal to focal length.
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)
4. Which of the following descriptions about a convex lens is/are correct?
- (1) A convex lens is a converging lens.
 - (2) The image formed by a convex lens is always real.
 - (3) The image formed by a convex lens is always magnified.
- A. (1) only
B. (2) only
C. (1) and (3) only
D. (1), (2) and (3)

A

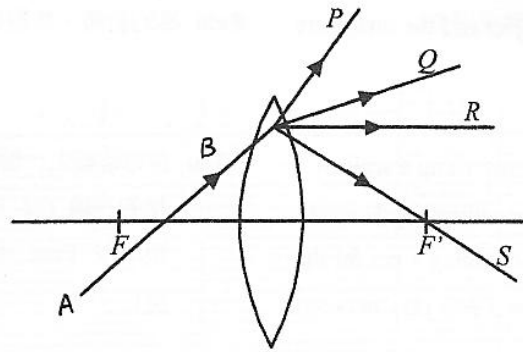
B

D

A

5. In the figure, F , F' are the foci of the convex lens and AB is an incident ray. Which of the following paths best represents the emergent ray?

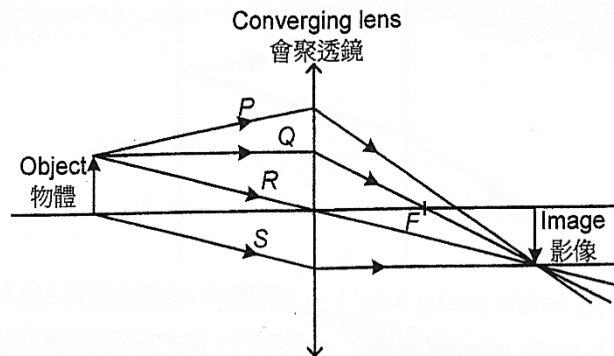
- A. P
B. Q
C. R
D. S



B

6. In the figure above, F is the focus of the converging lens. Which of the four rays in **INCORRECTLY** drawn?

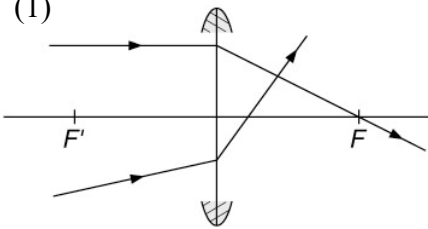
- A. P
B. Q
C. R
D. S



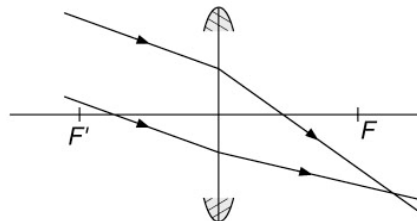
D

7. If F and F' are the foci of a convex lens, which of the following ray diagrams is/are **incorrect**?

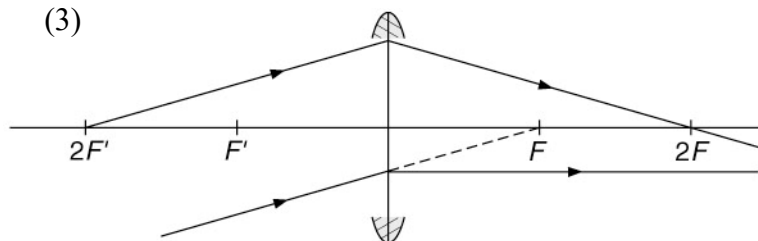
(1)



(2)



(3)

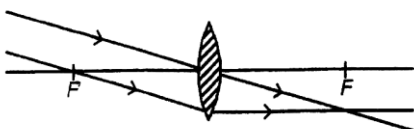


- A (3) only
B (1) and (2) only
C (2) and (3) only
D (1), (2) and (3)

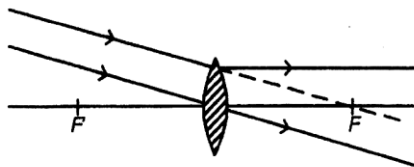
D

8. Which of the following ray diagrams is/are correct?

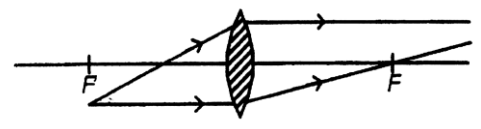
(1)



(2)



(3)

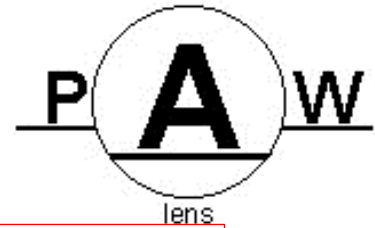


- A. (1) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only

A

9. Some letters are viewed through a lens. The image of the letters is shown in the following figure. Which of the following statements is/are correct?

- (1) The letters are located within one focal length from the lens.
(2) The image is virtual.
(3) The lens is a convex lens.



- A (1) only
B (3) only
C (1) and (2) only
D (1), (2) and (3)

(1): To produce a erect magnifying effect, the object must be placed within a distance of a focal length to the lens.
(2): Only a convex lens will produce a magnifying effect.

D

10. A distant object gradually approaches a convex lens along the principal axis. The image formed will
(1) remain inverted throughout
(2) change from real to virtual after the object passes through the principal focus.
(3) increase in size before the object reaches the principal focus.

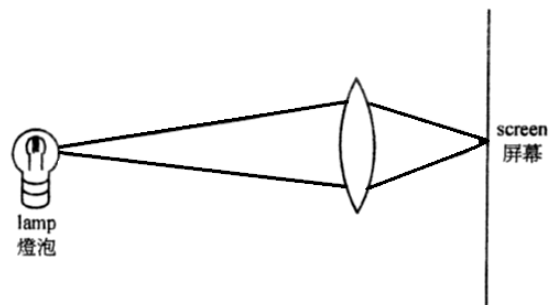
- A. (1) only
B. (2) only
C. (3) only
D. (2) and (3) only

D

11. A convex lens L is placed between a screen and a lamp as shown in the diagram. A sharp image is formed on the screen. When the lens L is moved a certain distance towards the lamp, a sharp image is formed again on the screen. Which of the following statements concerning the second image is/are correct?

- (1) The image is larger than the first image.
(2) The image is real.
(3) The image is inverted.

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (1), (2) and (3)

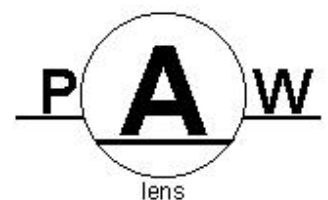


D

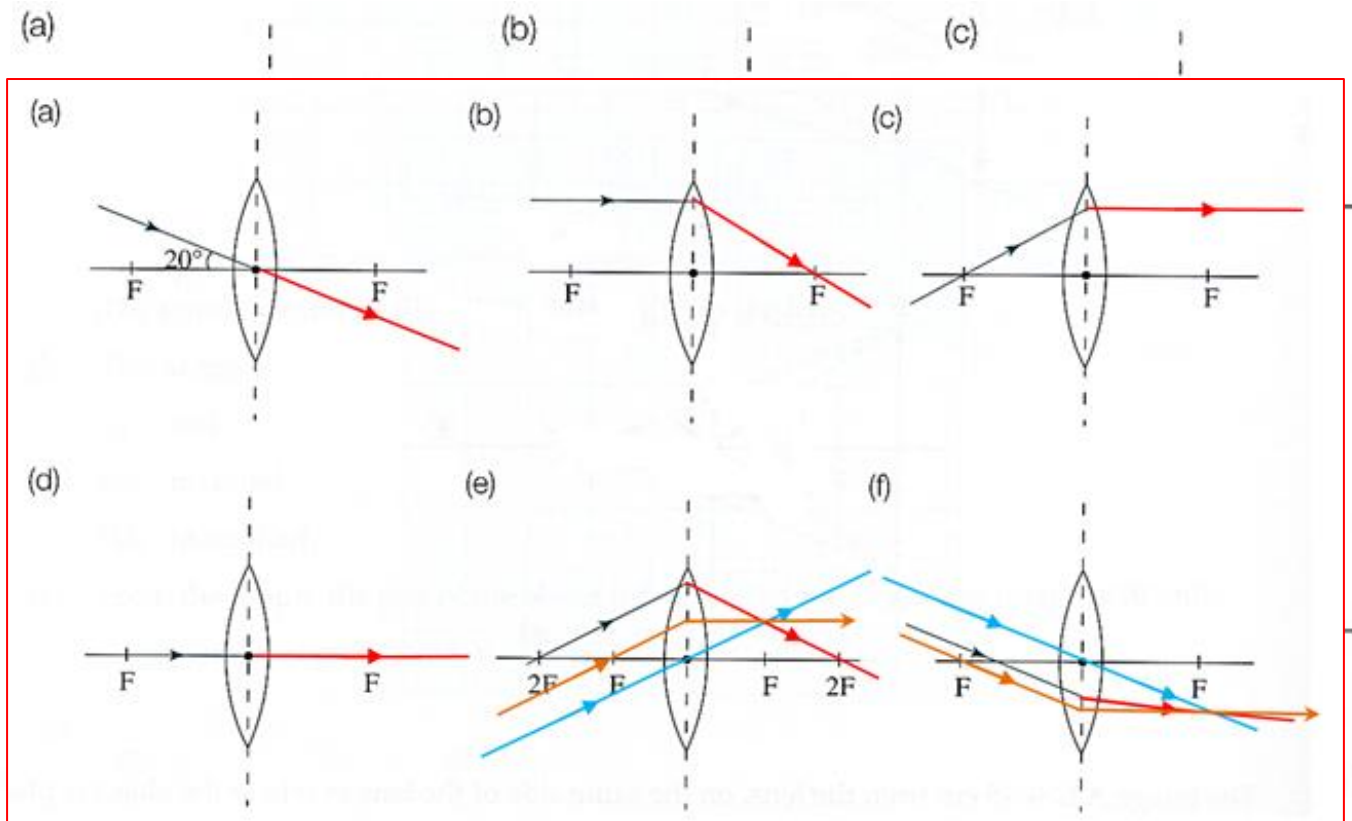
12. Some letters are viewed through a lens. The image of the letters is shown in the following figure. If the lens is moved away from the letters, which of the following correctly describes how the appearance of the image changes?

A

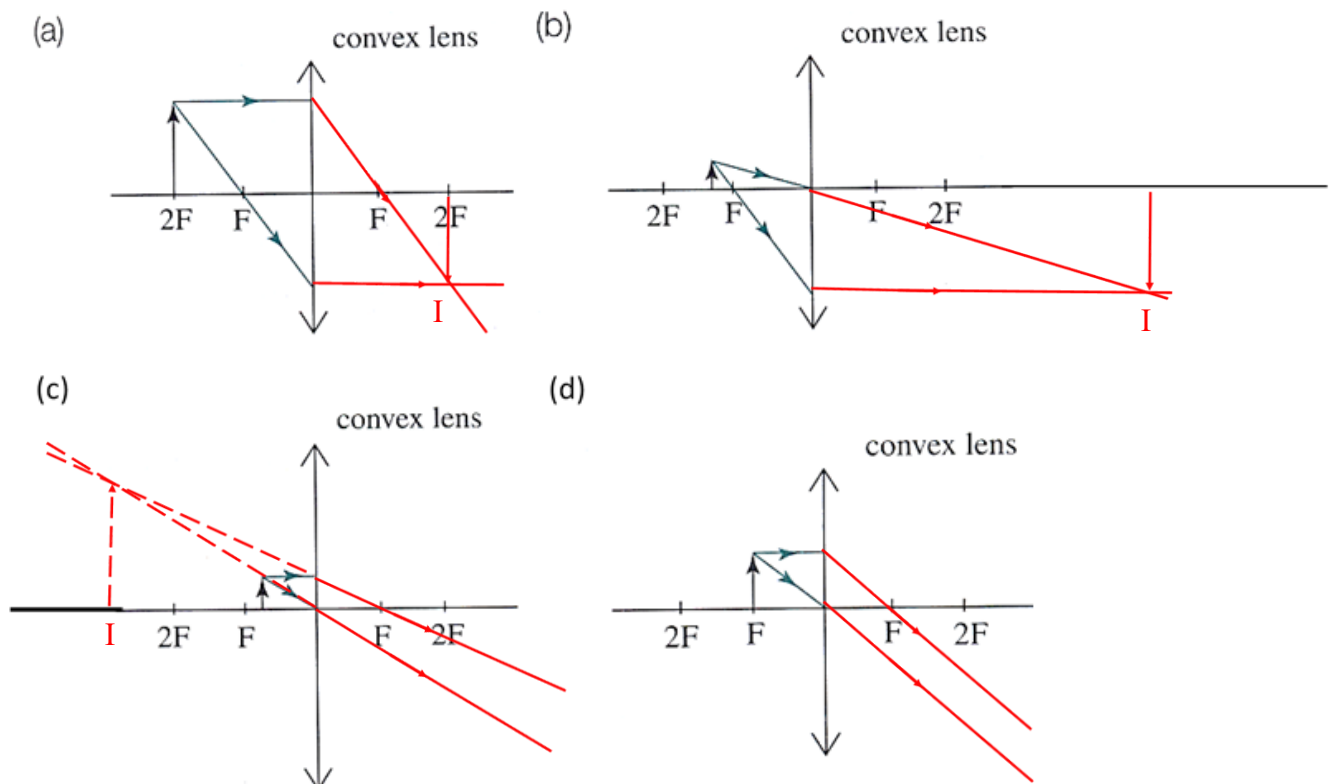
- A. Magnified, erect → magnified, inverted → diminished, inverted
B. Magnified, erect → magnified, inverted → more magnified, inverted
C. Magnified, erect → diminished, inverted → more diminished, inverted
D. Magnified, erect → magnified, inverted → more magnified, erect



13. Draw the refracted rays.



14. Draw the refracted rays and locate the image.



15. Victor views a pearl through a lens as shown. His image can be seen through the lens.

- (a) (i) State the nature of his image.
(ii) What kind of lens is this?
- (b) What should be the distance between the lens and the pearl? Explain briefly.

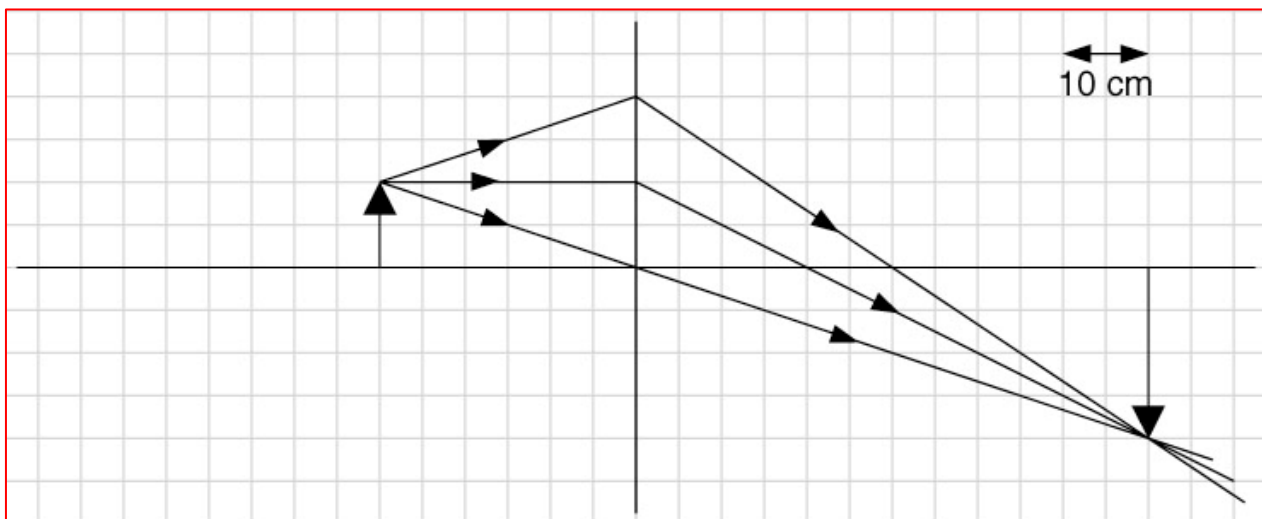


- (a) (i) virtual, erect, magnified
(ii) Convex lens
- (b) In order to produce a magnified image of the pearl, the distance between the pearl and the lens should be less than the focal length.

16. An object is placed in front of lens L. The following ray diagram shows that a light ray from the object is refracted by L.

- (a) What type of lens is L? Explain briefly.
- (b) By adding suitable light rays to the above diagram, find the image distance and the focal length of the lens.
- (c) State the nature of the image.

- (a) Convex lens because the ray converges.
- (b) Image distance = 60 cm
Focal length = 20 cm
- (c) Real, inverted, magnified



17. An object is placed in front of a converging lens as shown below. The four light rays shown come from the object.

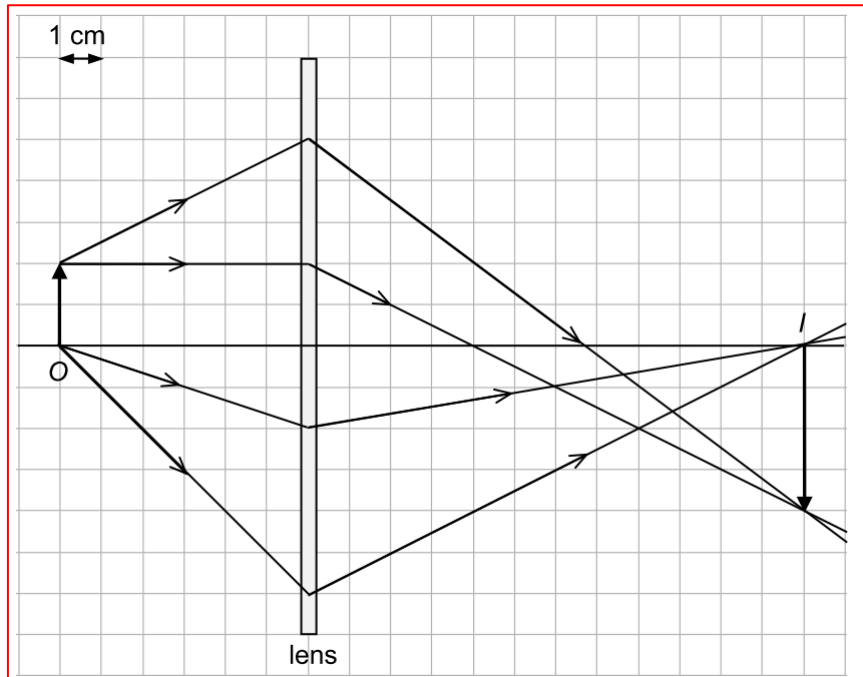
(a) (i) Complete the paths of the light rays in the figure and locate the image. (4 marks)

(ii) What is the focal length of the lens? (1 mark)

(b) Amy uses the lens to read a newspaper. A headline on the newspaper is at a distance of 1.5 cm from the lens. Can Amy use a screen to capture the image? Explain your answer. (2 marks)

(a) (ii) 4 cm

(b) No, because the image is virtual.



18. A letter 'F' illuminated by a lamp-housing is put in front of a convex lens as shown. The image of the letter is captured by a screen.

(a) Sketch the image formed on the screen. (1 mark)

(b) The lower half of the lens is then covered.

(i) Can the whole image be formed on the screen? Explain your answer with the aid of a ray diagram. (4 marks)

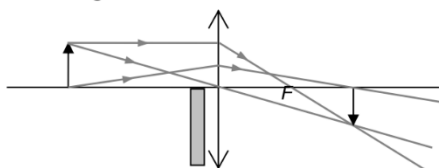
(ii) Does the brightness of the image change? Explain briefly. (2 marks)

(c) If the screen is moved a bit closer to the lens, the image will become blurry. Explain briefly. (1 mark)

(a)

(Correct image)

(b) (i) Yes, the whole image can be formed on the screen.



(Correct application of construction rules to locate the image)

(Ray from tail of object reaches tail of image via lens)

(ii) The image becomes dimmer as the amount of light forming the image decreases.

(c) Light from a point on the object will become a small patch instead of a point on the screen.

2 × 1A

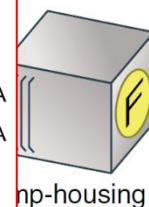
1A

1A

1A

1A

1A



convex lens

